Md Hasebul Hasan

Machine Learning Engineer | Ph.D. Researcher | LLM Systems & MLOps

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LinkedIn | GitHub | Google Scholar | Portfolio

Summary

Machine learning engineer with **3+ years of full-time experience at IQVIA**, building and deploying production ML/LLM systems. Current Ph.D. researcher focused on trustworthy and efficient AI. Production work contributed to a **15% sales increase**; research accepted at NeurIPS 2026 and published at ICML, EACL, and IEEE COMPSAC.

Technical Skills

Cloud / MLOps: AWS SageMaker, Bedrock, EC2/ECR, Kubernetes, Kubeflow, Docker, GitLab CI/CD, Git

ML / LLM: PyTorch, TensorFlow, Hugging Face, scikit-learn, LangChain, RAG, LLM fine-tuning and evaluation, agentic systems

Languages: Python, C/C++, Java, SQL, JavaScript

Data: Snowflake, PostgreSQL, MySQL, MongoDB

Research Experience

Graduate Research Assistant **Aug 2025 – Present**

University of Texas at Arlington – Arlington, TX *Ph.D. Researcher*

- Built **DeEscalWild**, a real-world LLM benchmark with **1,500 scenarios, 285K+ dialogue turns, and approximately 4.7M tokens**; first-author paper accepted to the NeurIPS 2026 Evaluations and Datasets Track.
- Led data curation, transcription, diarization, privacy-preserving anonymization, validation, SLM fine-tuning, and evaluation for the benchmark.
- Research **Llama-2-7B KV-cache compression on NVIDIA H100 GPUs**, profiling memory and latency and evaluating low-rank, token-retention, and fixed-rank coverage strategies for efficient inference.

Industry Experience

Machine Learning Engineer **May 2022 – Jul 2025**

IQVIA – Dhaka, Bangladesh *Full-time*

- Developed and deployed production ML/time-series systems with **Python, PyTorch, TensorFlow, and Snowflake**; solutions contributed to a **15% increase in sales**.
- Built time-series data pipelines over enterprise data marts, covering data engineering, preprocessing, model optimization, evaluation, and deployment.
- Architected **LangChain multi-agent/LLM workflows** and fine-tuned open-source LLMs for domain-specific applications.
- Productionized ML workloads using **AWS SageMaker/Bedrock, Kubernetes, and Kubeflow** across enterprise ML pipelines.
- Integrated **Docker, EC2/ECR, and GitLab CI/CD** into production ML workflows.
- Conducted **AI security red-team testing** for reliability, bias, and vulnerability assessment.

Selected Research & Publications

DeEscalWild **NeurIPS 2026, Evaluations & Datasets – Accepted**

A Real-World Benchmark for Automated De-Escalation Training with SLMs.

First author; created a 1,500-scenario real-world benchmark and led data curation, privacy-preserving anonymization, SLM fine-tuning, and evaluation.

MapAgent **EACL 2026 Findings**

A Hierarchical Agent for Geospatial Reasoning with Dynamic Map Tool Integration.

First author; designed and evaluated a hierarchical LLM agent with dynamic map-tool integration for compositional geospatial reasoning.

MapEval **ICML 2025 Spotlight – Top 2.6%**

A Map-Based Evaluation of Geo-Spatial Reasoning in Foundation Models.

Co-designed the task taxonomy and evaluation; contributed to dataset construction, annotation, and human verification.

GuardFL **IEEE COMPSAC 2024**

Detection and Removal of Malicious Clients Through Majority Consensus.

First author; developed a majority-consensus method for detecting and removing malicious clients in federated learning.

Education

University of Texas at Arlington **Aug 2025 – Present**
Ph.D. in Computer Science and Engineering *GPA: 4.00/4.00*
Expected graduation: June 2028.

Bangladesh University of Engineering and Technology **2022 – 2025**
M.Sc. in Computer Science and Engineering *GPA: 3.75/4.00*

Bangladesh University of Engineering and Technology **2017 – 2022**
B.Sc. in Computer Science and Engineering *GPA: 3.77/4.00; Rank: 31/127*

Teaching Experience

Lecturer, Computer Science & Engineering **Nov 2022 – Jul 2024**
Bangladesh University of Engineering and Technology (BUET) *Part-time*

- Taught six undergraduate lab/sessional courses and supervised **30 student projects** across algorithms, software engineering, databases, data structures, digital logic, and programming.

Awards & Grants

- **ICPC South Central Regional, Division 2 Champion (2026).** *Code Blazers*, The University of Texas at Arlington; November 7, 2026. Solved **9 of 13 problems**.
- **NeurIPS 2026 Financial Assistance Award.** Accommodation and registration for both the conference and workshops in Atlanta, GA.
- **UT Arlington Travel Grant (2026): US$500** to present *DeEscalWild* at NeurIPS 2026, Atlanta, GA.
- **Additional awards:** IQVIA Impact Platinum (2024), Silver (2025); University Merit Scholarship; Dean's List (3 years).